HYDRA: project description

HYDRA is a machine learning–based prototype designed to assess hydration status and identify possible dehydration. It uses U-Net, a deep learning model for image segmentation, to isolate relevant regions in eye images for further analysis. Image-derived features are combined with heart rate and oxygen saturation (SpO_2) measurements collected through a MAX30102 sensor. A Support Vector Machine (SVM) classifier analyzes the extracted features to distinguish hydration categories. By integrating image segmentation, physiological sensing, and classification, HYDRA investigates a multimodal approach to hydration assessment.

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